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Additional malignancies and mortality in uveal melanoma: A 20-year follow-up of a Norwegian patient cohort

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Abstract
Purpose: The purpose of this study is to explore the frequency of additional primary malignancies in uveal melanoma (UM) patients and cause-specific mortality, to help guide surveillance strategies after UM.
Methods: All patients diagnosed with UM at Oslo University Hospital during 1990–2017 were eligible for inclusion. Linkage to the Cancer Registry of Norway obtained information on additional malignancies and cause of death throughout 2019. UM patients were categorized according to timing of additional malignancy (prior/simultaneously or after UM) or no additional cancer, and by UM stage at diagnosis. Age-adjusted mortality rates were presented per 1000 person-years with 95% confidence intervals (CI).
Results: The study population included 960 UM patients: 77% were diagnosed in stage and I/II and 56% were men. Mean age at diagnosis was 63 years. Additional malignancies were observed in 152 patients prior/simultaneous to UM, and in 120 patients >1 year after UM. Overall, mortality per 1000 person-years was 3.5 (95% CI 3.1–3.9) for UM and 3.0 (2.6–3.4) for other causes. Lowest UM mortality [1.3 (0.60–2.1)] was seen in patients with a second malignancy after UM, regardless of stage. Highest UM mortality was seen for UM patients in stage III/IV, both without [16.1 (13.2–19.1)] and with any additional malignancy [16.9 (6.6–27.3)].
Conclusion: Our results support that UM patients frequently have additional malignancies, both before and after UM. Low-UM mortality in patients with a primary malignancy after UM, might indicate less aggressive UM. The cumulative UM mortality flattens about 10 years after diagnosis and annual follow-up of patients for 10 years seems adequate.

KEYWORDS
additional primary malignancy, long-term follow-up, mortality, stage, uveal melanoma

1 | INTRODUCTION

Ocular melanoma is the second most common subtype of melanoma, accounting for about 5% of all melanoma diagnoses (Chang et al., 1998), and uveal melanoma (UM) is the most common primary intraocular malignancy in adults (Ghazawi et al., 2019). In Europe, UM incidence increases from south to north, with highest rates in the Nordic countries with >8 per million (Virgili et al., 2007). The American Joint Committee on Cancer (AJCC) staging system classify tumour stage (T1–4) based on basal diameter and tumour thickness, location, and extraocular extension (Amin et al., 2017). The incidence rate has been stable for medium and large tumours (T3–T4) but has increased for those smaller (T1–T2; Smidt-Nielsen et al., 2021).

Follow-up studies have reported multiple malignancies in patients with UM, but the frequency varies (Bagger et al., 2022; Bergman et al., 2006; Callejo et al., 2004; Diener-West et al., 2005; Heussen et al., 2016; Laíns et al., 2016). Recently, a Danish nationwide cohort study showed a 21% increased incidence of a new primary cancer following a posterior UM, although not restricted to specific cancer forms (Bagger et al., 2022). Screening for metastases at diagnosis of primary UM has shown that about 3% had simultaneously another cancer (Freton et al., 2012). Surveillance may thus reveal malignancies that may not have been detected without

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extensive examination. A reason for multiple malignancies may be hosting of mutations. It is documented that a germline BAP1 mutation can cause multiple malignancies (e.g., dysplastic nevus syndrome, cutaneous melanoma, and renal cell carcinoma), and it has been suggested that this is also associated with tumours in other organs (e.g., pancreas, breast, lung, and meninges; Abdel-Rahman et al., 2011). Moreover, cancer treatment may cause a new primary malignancy. In bilateral retinoblastoma, secondary malignancies are caused by mutations in the retinoblastoma gene, and radiation therapy has been documented to induce additional second cancers with a relationship to dose and mode of delivery (Wong et al., 1997). Based on the Collaborative Ocular Melanoma Study, however, it was concluded that radiotherapy in UM patients does not increase development of second primary cancers (Diener-West et al., 2005).

A Danish nationwide study over a 70-year period showed that UM specific mortality decreased over time for all tumour sizes except T4, though this was most prominent for smaller (T1–2) tumours (Smidt-Nielsen et al., 2021). Decreasing tumour size at diagnosis and decreasing UM mortality from 1960 to 2010 was also recently reported from Sweden (Gill et al., 2022). Occurrence of additional malignancies has been shown to influence mortality rates, suggesting that patients with UM as their first primary cancer have better UM-specific survival and overall survival than patients with a non-UM as their first primary cancer (Andreoli et al., 2015).

We aimed to explore the frequency of additional primary malignancies prior/simultaneously to or after a diagnosis of UM, and to examine mortality from UM and other causes, based on a large cohort of UM patients linked to the Cancer registry of Norway (CRN), to help guide surveillance strategies after UM.

2 | METHODS

2.1 | Study population

Since 1990, all patients with a clinical diagnosis of a UM have been included in a UM database at the Eye Department at Oslo University Hospital, which is one of two hospitals in Norway with responsibility for UM patients. The establishment of this database aimed to observe the results of therapy, explore different modalities, and collect clinical data important for medical surveillance. A previous cancer diagnosis was not a reason for exclusion in the database. To minimize diagnostic errors and treatment interference, UM diagnoses were verified from biopsies and pathological examinations. Patients had annual medical controls for 10 years. Thereafter, data collection was based on reports from Ophthalmologists, home hospital, and telephone calls to patients or relatives.

The present study was based on an audit of this database. The Regional Committee for Medical Research Ethics Norway (REK) approved this study (REK South-East, 2017/705). This study has been conducted in accordance with the tenets of the Declaration of Helsinki. The UM patients alive were asked to participate via an information letter or telephone call, and signed consent was required before enrolment. Deceased patients were exempt from providing consent. Patient data were validated based on clinical journals and data from the Cancer Registry of Norway (CRN) and the Cause of Death Registry of Norway (CDRN). The CRN has recorded information on all malignant diagnoses in Norway since 1953. Several independent sources ensure complete and high-quality data (Larsen et al., 2009). Cancer diagnoses are recorded using the International Classification of Diseases (ICD) 7th edition, ICD of oncology 2nd and 3rd edition (ICD-O-2 and ICD-O-3) and converted to the ICD 10th revision (Cancer Registry of Norway, 2022). The UM patient cohort was linked to the CRN and CDRN using the 11-digit identification number assigned to all Norwegian residents to obtain information about all malignant diagnoses (location, clinical stage, date of diagnosis, age at diagnosis) prior/simultaneously to and after the UM diagnosis, and about date and cause of death, until 31 December 2019.

Based on information from clinical records, UM stage was classified according to the AJCC, 7th Edition (Edge & Compton, 2010), and primary tumour site was classified based on tissue involvement before treatment (choroid, ciliary body, iris and unspecified). Data from the CRN allowed tracking of all additional primary malignancies in the UM patients, and they were categorized as: UM with no additional malignancy, UM with additional malignancy prior/simultaneously to (±1 year) UM diagnosis, and UM with additional malignancy after (>1 year) UM diagnosis. Cutaneous basal cell carcinomas were excluded. Clinical and histopathological materials available up to the time of death were routinely retrieved from clinical departments and general practitioners. These data were used to verify the cause of death from the CRN and allowed for a distinction between death from UM and death from other causes.

The study population was based on all patients included in the database during the diagnosis period 1990–2017 ($n = 1017$). We excluded patients without Norwegian identification number ($n = 5$), those not receiving treatment ($n = 28$), denied study participation ($n = 9$), other diagnoses than UM and benign tumours ($n = 15$), leaving a study population of 960 UM patients.

2.2 | Statistical analyses

Descriptive statistics were presented, as the frequencies and proportions for categorical variables and means with ranges for continuous variables. Description of additional malignancies was presented using ICD-10 codes and stratified according to the time of the additional diagnosis (prior/simultaneously to or after UM diagnosis).

To calculate mortality rates, patients were followed from date of UM diagnosis to date of death, emigration, or end of follow-up on 31 December 2019, whichever occurred first. We examined risk of death separately for deaths due to UM and other causes, and by timing of additional malignancy (prior/simultaneously or after UM) and AJCC stage at UM diagnosis. Age-adjusted mortality rates were presented per 1000 person-years with 95% confidence intervals. Age-adjustment was performed through direct standardization based on the overall age-distribution (≤49, 50–59, 60–69, 70–79, ≥80) in the cohort.

Stacked cumulative mortality rates were presented graphically to illustrate cause-specific mortality from the time of diagnosis with up to 20 years of follow-up for the whole cohort (based on observed events), and by timing of additional malignancy (none, prior/simultaneously to UM, after UM) based on flexible parametric survival models (Royston & Parmar, 2002). Cumulative number of deaths and patients alive are presented at every 5-year time-point during 20 years of follow-up. Further, to ease the comparison between groups, also non-stacked cumulative mortality rates were presented by timing of additional malignancy, separately for deaths due to UM and other causes, and AJCC stage at UM diagnosis. STATA/SE (version 17.0) was used for statistical analyses. The STROBE guidelines for reporting of observational studies were followed (von Elm et al., 2008).

3 | RESULTS

The study cohort had a predominance of men (56%), the mean age at diagnosis was 63 years (range 10–95), and 77% of cases were diagnosed in 2000 or later (Table 1). Most (69%) UM was in the choroid, 24% involved the ciliary body, and 6.3% was in the iris. A quarter (26%) of patients were diagnosed in stage I, while 52%, 21%, and 1.5% were in stage II, III, and IV, respectively. Patients were primarily treated with radiation (71%) and enucleation (22%). By end of 2019, 48% were still alive, 28% had died due to UM, and 24% had died due to other causes.

Overall, 333 additional malignancies were diagnosed among 272 UM patients. A total of 197 (n = 197) cancers were diagnosed in 152 patients prior/simultaneously (±1 year) to UM diagnosis, and 136 were diagnosed in 120 patients >1 year after UM (Table 2). Cancers of breast (n = 47), prostate (n = 53), colorectum (n = 38), and urinary organs (n = 30) were the most common additional malignancies overall. The only malignancy that was most frequent after UM was lung cancer.

In UM patients diagnosed with stage I/II disease, 109 patients had additional cancer prior/simultaneously to UM, and 106 patients had additional cancer after UM (Table 3). For stage III/IV, most patients had the additional cancer prior/simultaneously to UM (n = 43), rarely after (n = 14). The mean age at UM diagnosis was quite similar for stage I/II [62.5 years (SD 15.6)] and III/IV [65.9 years (SD 12.8)] patients.

The median time from UM diagnosis to end of follow-up was 8.1 (range 0.003–29.9) years. Overall, the UM mortality rate [deaths per 1000 (with 95% CIs)] was slightly higher [3.5 (3.1–3.9)] than from other causes [3.0 (2.6–3.4)] (Table 3). Patients without additional cancer had higher UM mortality [4.2 (3.6–4.7)], than from other causes [2.3 (2.5–3.5)], while in patients with any additional cancer the mortality from other causes was the highest [5.3 (4.3–6.3) vs 2.7 (2.0–3.3)]. Low-UM mortality was seen in patients with additional cancers after UM, while high-UM mortality was seen in patients with additional malignancy prior/simultaneously to UM, both within stage I/II and III/IV UMs. The highest UM mortality was seen for stage III/IV UM, regardless of additional malignancy. Mortality from other causes was lowest in patients without additional malignancy [all

TABLE 1 Patient characteristics of the uveal melanoma patient cohort, n = 960.

	Number	Percentage	Mean (range)
Sex			
Women	425	44.3	
Men	535	55.7	
Year of birth	960		1943 (1906–1997)
Age at diagnosis, years	960		63.3 (10.5–94.9)
<=49	173	18.0	
50–59	193	20.1	
60–69	235	24.5	
70–79	240	25.0	
80+	119	12.4	
Period of diagnosis			
1990–1999	217	22.6	
2000–2009	393	40.9	
2010–2017	350	36.5	
Primary tumour site			
Choroid	665	69.3	
Ciliary body[a]	234	24.4	
Iris	60	6.3	
Unspecified	1	0.1	
Stage[b]			
I	245	25.5	
IIA	330	34.4	
IIB	168	17.5	
IIIA	136	14.2	
IIIB	58	6.0	
IIIC	8	0.8	
IV	14	1.5	
Unspecified	1	0.1	
Treatment			
Enucleation	208	21.7	
Radiation therapy	683	71.2	
Local resection	65	6.8	
Other[c]	4	0.4	
Status (31.12.2019)			
Alive	462	48.1	
Death, uveal melanoma	265	27.6	
Death, other causes	232	24.2	
Emigration	1	0.1	

[a]All tumours with involvement of ciliary body are included.
[b]According to AJCC staging system for malignant tumours, 7th edition.
[c]Cyclo-kryo (n = 1), endo-resection (n = 2), and transpupillary-thermo-therapy (n = 1).

TABLE 2 Numbers (n) of additional malignancies diagnosed prior/simultaneously (±1 year) or subsequent (>1 year) to the diagnosis of uveal melanoma (UM).

		Additional malignancies[a]		
		All	Prior/simultaneously to UM	After UM (>1 year)
ICD-10	Cancer site	*n*	*n*	*n*
C00-96	All sites[b]	333	197	136
C00-14	Mouth, pharynx	4	4	0
C18-20	Colorectum	38	18	20
C22-24	Liver, gallbladder, bile duct	3	1	2
C25	Pancreas	3	1	2
C15-17, C21, C26	Other digestive sites	9	7	2
C33-34	Lung, trachea	22	6	16
C30-32, C38	Other respiratory sites	5	3	2
C43	Skin, melanoma	25	19	6
C44	Skin, non-melanoma[b]	14	6	8
C48-49	Soft tissue	1	0	1
C50	Breast	47	31	16
C51-58	Female genital organs	21	14	7
C61	Prostate	53	35	18
C60, C62-63	Male genital organs	3	3	0
C64-68	Urinary organs	30	14	16
C69	Eye[c]	3	2	1
C70-72	Central nervous system[d]	19	14	5
C37, C73-75	Endocrine glands	6	2	4
C81-96	Lymphoid/haematopoietic tissues	23	16	7
C39, C76, C80	Other and unspecified sites	4	1	3

[a]Diagnosed in 272 UM patients during 1954–2019: 222 patients had one additional malignancy, 39 patients had two additional malignancies, and 11 patients had three or more additional malignancies.

[b]Basal cell carcinomas are not included.

[c]Specifications of the three cases; Case 1: Primary UM in right eye in 2017, with a *prior* primary conjunctival melanoma in 2012. Case 2: Two primary UM simultaneously in right and left eyes. Case 3: Primary UM in right eye in 2009 with a *subsequent* primary UM in the same eye in 2017.

[d]Meningioma included.

stages, 2.3 (2.5–3.5), and highest for stage III/IV UM with any additional malignancy, 11.8 (1.8–21.9)].

Figure 1 illustrates the stacked cumulative cause-specific mortality overall and by categories of additional malignancy, while Figures 2 and 3 presents the non-stacked cumulative mortality from UM diagnosis up to 20 years of follow-up. The mortality from UM increased rapidly the first 5 years after diagnosis. It continued to increase with a less steep intensity until about 10 years after diagnosis, after which it flattened (Figure 1a). Further, the cumulative UM mortality was 20%, 29%, 33%, and 35%, respectively, after 5, 10, 15, and 20 years from UM diagnosis. The cumulative mortality from other causes increased linearly throughout the first 20 years of follow-up.

Patients with additional malignancy after UM seemed to have the lowest UM mortality, while UM mortality increased with the same shape and magnitude for patients with additional cancer prior/simultaneously to UM as for those without additional cancer (Figures 1b–d and 2a). A similar pattern was seen for stage I/II and stage III/IV patients, although the level of mortality differed by stage (Figure 3a,c). The cumulative mortality from other causes was lowest for patients having no additional cancer and highest for patients with additional cancer prior/simultaneously to UM (Figures 1b–d and 2b). This pattern was also similar for stage I/II patients (Figure 3b). For stage III/IV patients, the highest mortality from other causes was seen for patients with additional malignancy after UM (Figure 3d).

4 | DISCUSSION

In this cohort of 960 UM patients, we found that 28% had at least one additional malignancy, prior/ simultaneously to or >1 year after their UM diagnosis. UM patients diagnosed in stage I/II had a similar frequency of additional cancer prior/simultaneously to and after UM, while stage III/IV patients mainly had additional cancer prior/simultaneously to UM. Low-UM mortality was seen in patients with additional cancers after UM. The highest UM mortality was seen in stage III/IV UM patients. The UM mortality increased rapidly the first 5 years after diagnosis and thereafter flattened, while other cause mortality increased linearly throughout 20-year of follow-up.

Our cohort consisted predominantly of men, with a slightly higher proportion than reported from population-based data from US (Laíns et al., 2016), Sweden (Bergman

TABLE 3 Number (*n*) of uveal melanoma (UM) patients without and with additional primary malignancy, diagnosed prior or simultaneously (±1 year), or after UM diagnosis, and deaths and age-adjusted mortality rates with 95% confidence intervals (CI) for UM and other causes, by stage.

	n	Pyr[a]	UM deaths, *n*	Um, Mortality rate per 1000 pyr (95% CI)[b]	Other causes deaths, *n*	Other causes, Mortality rate per 1000 pyr (95% CI)[b]
All stages	960	7798	272	3.5 (3.1–3.9)	235	3.0 (2.6–3.4)
No additional malignancy	688	5725	213	4.2 (3.6–4.7)	125	2.3 (2.5–3.5)
Any additional malignancy	272	2073	59	2.7 (2.0–3.3)	110	5.3 (4.3–6.3)
Prior or simultaneous to UM	152	767	41	5.1 (3.5–6.8)	56	5.7 (4.1–7.2)
At least one after UM (>1 year)	120	1306	18	1.3 (0.60–2.1)	54	4.8 (3.3–6.4)
Stage[c] I–II						
No additional malignancy	528	4905	104	2.4 (1.9–2.8)	102	3.0 (2.4–3.5)
Any additional malignancy	215	1830	30	1.5 (0.97–2.1)	96	5.1 (4.7–6.1)
Prior or simultaneous to UM	109	638	15	2.6 (1.1–5.0)	48	5.7 (4.2–7.4)
At least one after UM (>1 year)	106	1190	15	1.0 (0.50–1.5)	48	4.8 (3.2–6.5)
Stage[c] III–IV						
No additional malignancy	159	810	108	16.1 (13.2–19.1)	23	2.8 (1.6–4.0)
Any additional malignancy	57	246	29	16.9 (6.6–27.3)	14	11.8 (1.8–21.9)
Prior or simultaneous to UM	43	129	26	23.1 (11.8–34.3)	8	10.3 (0.28–20.3)
At least one after UM (>1 year)	14	116	3	3.8 (–)[d]	6	14.4 (–)[d]

[a]Pyr; person-years of follow-up from diagnosis of uveal melanoma to death.
[b]Age-adjusted based on the age distribution (≤49, 50–59, 60–69, 70–79, ≥80) in the total cohort.
[c]Based on the AJCC staging system for malignant tumours, 7th edition.
[d]95% CI not presented due to too few deaths and person-years of follow-up.

et al., 2002), and Denmark (Bagger et al., 2022). We have no explanation for this difference. The proportion involving the ciliary body was also high, compared to other studies, that might result from variation in criteria for inclusion in this category. Further, few UMs were diagnosed in the first period of diagnosis (1990–1999). Centralisation of UM treatment in the late 1990s to two centres in Norway, Oslo University Hospital and Haukeland University Hospital in Bergen, may explain the inclusion of more cases in the database from 2000 onwards.

In the Swedish Cancer Registry, 5% of UM patients were registered with additional cancer prior to UM (Bergman et al., 2006). Similar proportions (4%–7%) have been reported from small, hospital-based cohorts (Callejo et al., 2004; Heussen et al., 2016). In the present study, the proportion was 16%; however, new primary cancers in the same year, as the UM (simultaneously), were also included. Further, we found that 13% of UM patients had malignant diagnosis after UM. This is a slightly higher proportion than was reported with 10-year follow-up from the COMS trial (9.6%; Diener-West et al., 2005). However, that study excluded all non-melanoma skin cancer, while we excluded only basal cell carcinomas. Further, our cases were followed for up to 20 years from UM diagnosis, which might have led to increased detection of additional cancers. In the SEER database, 16% of UM patients were found to have a subsequent cancer (Laíns et al., 2016). In that study, UM cases were included from 1973, giving more than 30 years of follow-up. However, compared to the reference population, the excess cancer risk was 11%. A Swedish population-based study found a 13% elevated risk of subsequent cancer when comparing UM patients with the general population (Bergman et al., 2006), and, in Denmark, a 21% elevated cancer risk following UM has been reported (Bagger et al., 2022). In comparison, a population-based U.S. study found that 8.1% of cancer survivors from the 10 most common cancers developed a second primary malignancy (Donin et al., 2016). Compared with non-cancer controls, cancer survivors undergo more screening examinations during surveillance (Corkum et al., 2013) and knowledge about malignant comorbidity in UM patients may increase awareness in medical surveillance after UM that can reveal additional malignancies. To date, however, there are no evidence-based guidelines for surveillance of new primary cancer during follow-up of UM patients (Bagger et al., 2022).

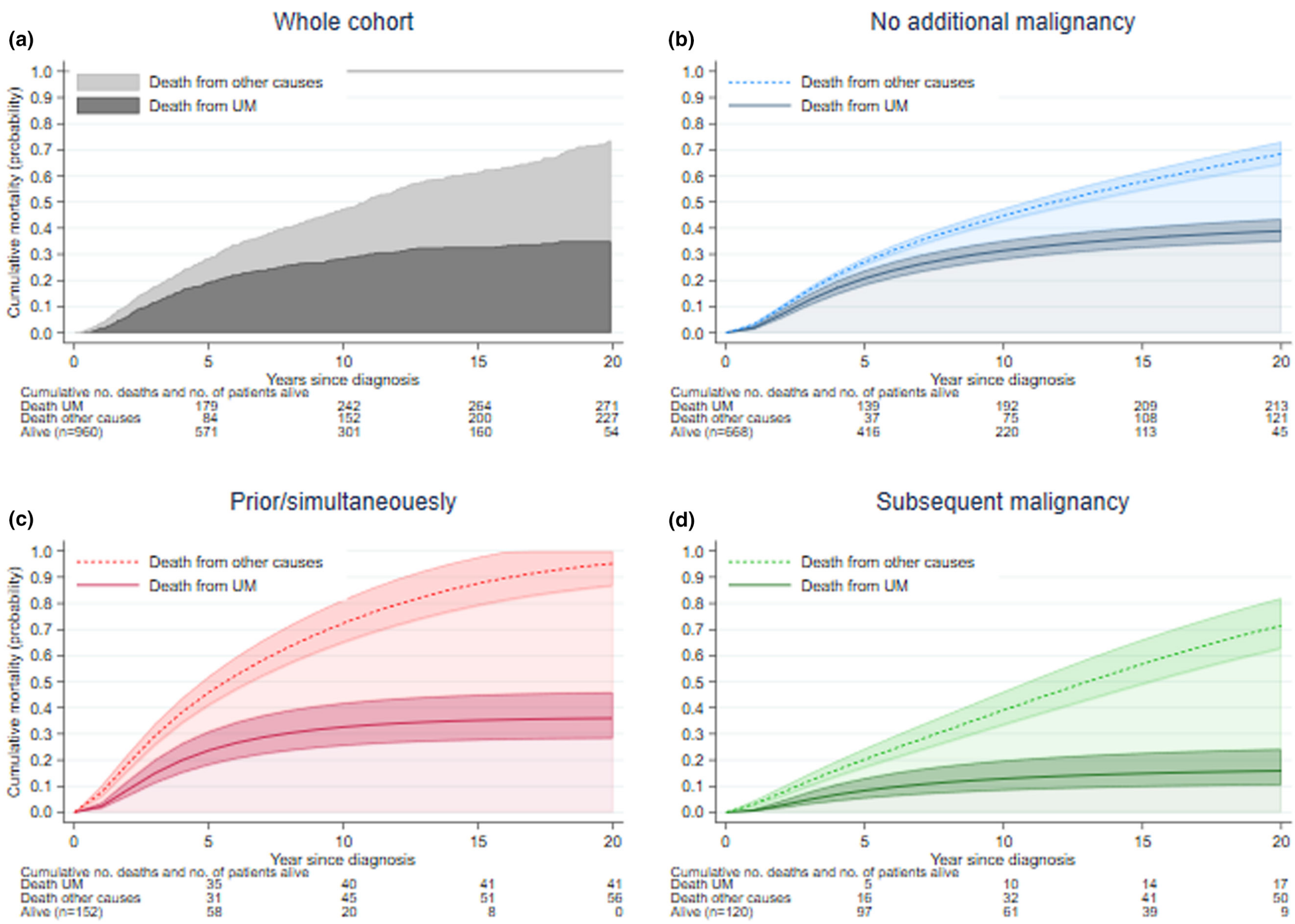

FIGURE 1 Stacked cumulative mortality for uveal melanoma (UM) death and other causes, from UM diagnosis up to 20 years of follow-up, for (a) all UM patients, and in UM patients with (b) no additional malignancy, (c) additional malignancy prior/simultaneously to UM, and (d) additional malignancy subsequent to UM, with 95% confidence intervals. Cumulative number of deaths and patients alive are presented at every 5-year time-point during 20 years of follow-up.

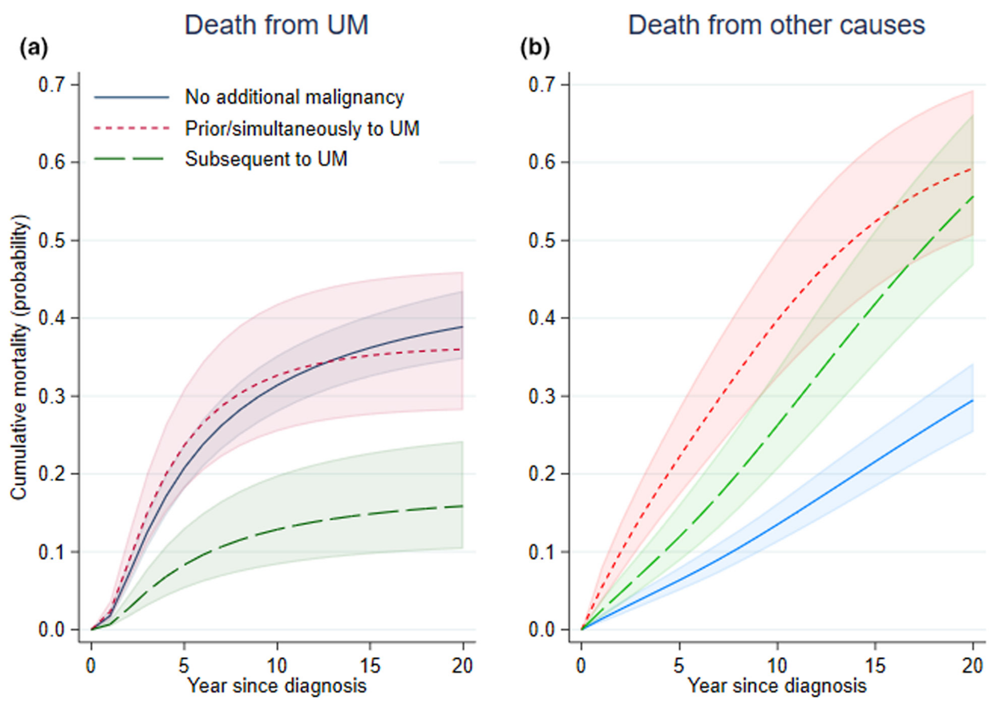

FIGURE 2 Cumulative mortality with 95% confidence intervals (shaded area) for (a) uveal melanoma (UM) and (b) other causes, by additional malignancy (no, prior/simultaneously or subsequently to UM), for all UM patients.

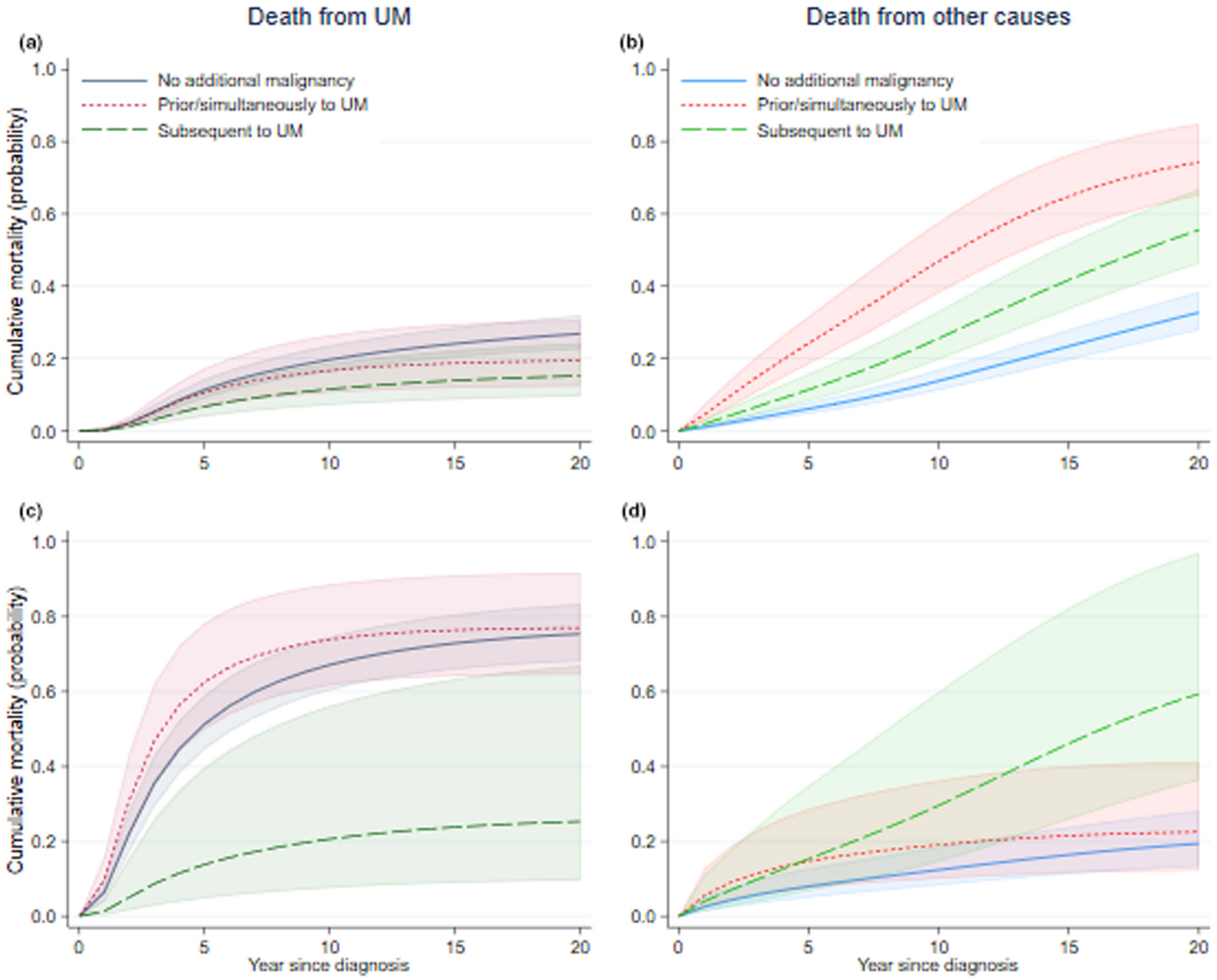

FIGURE 3 Cumulative mortality with 95% confidence intervals (shaded area) for uveal melanoma (UM) and other causes by additional malignancy (no, prior/simultaneously or subsequently to UM), for UMs in stage I/II (a, b) and stage III/IV (c, d).

Although neither the Swedish nor the Danish study found significantly increased risk for any specific cancer type, the most frequent additional cancers were of breast, prostate, colorectum, lung and skin, which also was seen in our data. As these are cancers with the highest incidence in western countries, including in Norway (Cancer Registry of Norway, 2022), this finding was expected. In our cohort, cancer of urinary organs was also frequent, both before and after UM, which indicates that this cannot only result from treatment or surveillance after UM (Table 2). Results from the SEER database showed an excess rate of kidney cancer after UM, suggested to result from inherited cancer predisposition (Laíns et al., 2016). A germline *BAP1* mutation can result in multiple malignancies (Abdel-Rahman et al., 2020), and this can indeed include UM and renal cell carcinoma (Battaglia, 2014). Testing UM tumours for somatic mutations is an important routine for prognostication (Dogrusöz & Jager, 2018; Shain et al., 2019), but testing for germline mutations is indicated only in special cases and was not performed in our material. Nonetheless, the frequency of such mutations is low (2%; Turunen et al., 2016), and germline *BAP1* mutation is not a likely explanation for the high frequency of additional cancer in our UM population.

We found that patients with additional malignancies after UM had the lowest UM mortality. These malignancies primarily occurred among UM patients in stage I/II. Our results were in line with findings reported from the SEER database, where UM patients with additional cancers had better overall and disease-specific survival, if their first cancer was a UM, compared to patients with a non-UM first cancer, though the effect depended on stage (Andreoli et al., 2015). Our data showed quite similar mean age at UM diagnosis for stage I/II and III/IV patients. Age was also considered in the analysis and can hardly explain differences in UM mortality between the groups. A likely explanation, however, might be less aggressive UM tumours that allow patients to live longer with the risk of developing a new malignancy. Lung cancer was the only malignancy that was more frequent after UM (16 cases vs. six cases prior/ simultaneously to UM), which also may contribute to reduced UM mortality in patients with additional cancer after UM.

Compared to the recently published study from Denmark (Bagger et al., 2022), our material had a higher proportion of patients in stage I (26% vs. 17%), and lower proportions in stage III/IV (23% vs. 27%) and unspecified (0.1% vs. 4%). Inclusion to our patient cohort started two decades later than the Danish study. Over time, the diagnostic modalities have improved and increased the ability to detect smaller tumours. Moreover, there has been a tendency to also treat smaller UMs without histopathological verification. These are likely explanations for the observed stage differences. Furthermore, increasing treatment of eye diseases, especially cataract and exudative aged macular disease, has increased eye consultations, which has increased the detection of more asymptomatic UM. Increasing incidence of stage I/II UM, unchanged incidence of stage III/IV, and decreasing tumour size has been observed in Denmark (Smidt-Nielsen et al., 2021). Concurrently, survival has improved (except for T4 cancers), and this improvement was most prominent for small tumours. Our study found the highest UM mortality in patients in stages III/IV, both among those with and without additional malignancies, as expected, except for patients with subsequent malignancy. However, the sample size was small for this group ($n = 14$).

Important strengths of our study are the population-based patient cohort of UM patients, recruited and followed over a long period, and the high-quality data. The UM database has been operated by only one person (NAE) since establishment. This minimizes the risk for inconsistent evaluation of the same phenomena over the study period. Additionally, the database was a part of the clinical evaluation at diagnosis and during follow-up, and results were recorded prospectively.

There are limitations to be considered when interpreting the study results. First, this is a descriptive study, and the UM database was not created for this study, but rather as a 10-year follow-up tool for UM patients after treatment. Data on of lifestyle factors, such as smoking, were therefore not included. The importance of *BAP1* and gene expression for prognosis was not clear at establishment of the database and was also not included. The trends observed in our study, however, are like the two other Nordic studies from Sweden (Bergman et al., 2006) and Denmark (Bagger et al., 2022). Second, although the CRN data are of high quality, there are shortcomings caused by lack of clinical reporting to the CRN. All malignant cells detected in tissue or cytological specimens are automatically reported to the CRN via pathology reports. However, if the diagnosis is based on clinical or imaging modalities only, the diagnosis must be manually reported by the doctor. This potential under-registration has been corrected, as the present material includes all patient information. Finally, we emphasize that our cohort includes iris melanoma, which has substantially better prognosis than other eye melanomas. However, these were few (6%) and may not substantially influence the mortality results.

In conclusion, our results support that UM patients frequently have additional malignancies. These malignancies were equally distributed prior/simultaneously and after UM in stage I/II patients, in stage III/IV patients, they were mainly prior/simultaneously to UM. A lower UM mortality in patients with additional malignancy after UM might indicate less aggressive UM, as advanced stage UM was the strongest predictor of UM mortality. The flattening curve in cumulative UM mortality 10 years after diagnosis supports that the current 10-year annual surveillance from diagnosis is adequate.

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DATA AVAILABILITY STATEMENT

The data underlying this article cannot be shared publicly due to patient privacy. Anonymous data can be shared for research purposes on reasonable request but will be subject to formal approval from the data owner (Oslo University Hospital, Ullevål) and Regional Committees for Medical and Health Research Ethics, according to the General Data Protection Regulation. Requests for data sharing should be directed to the corresponding author.

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